Songhyun Bae

Security Researcher

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Profile

I'm a Cyber Defense student at Korea University. I've been exploring security through hands-on vulnerability research across different systems and technologies. My current interests include fuzzing, kernel security, Web3 security, AI for security, and AI agent security.

Education

Korea University Mar. 2024 – Present
B.S. in Cyber Defense

Daejeon Daeshin High School Mar. 2021 – Feb. 2024
High School Diploma

Work Experience

HashLock — *Freelance Smart Contract Auditor* Aug. 2025 – Apr. 2026

- Audited Solidity, Vyper, and other Web3 smart contracts for high-severity vulnerabilities.

Activities

Web3 Audit Contests — *Participant* Dec. 2024 – Present

- 40 valid findings across public contests (2 Critical · 18 High · 19 Medium · 1 Low).
- Top placements: 1st Strobe SC (HackenProof), 1st Neutrl Protocol (Sherlock), 3rd Coinbase EIP-7702 Proxy (Cantina), 5th RAAC Core (CodeHawks).
- Full record: Audit portfolio.

CyKor — *Member* Sep. 2024 – Present

- Hacking club at Korea University.
- Organizer & challenge author, **CyKor CTF 2025** (Dec. 2025) — authored the Web3 challenge "Multi Chain Vault".
- Organizer & challenge author, **CyKor 2025 Spring Recruit CTF** (Mar. 2025) — authored two Web3 challenges [chal-1] [chal-2].

KISA Hunting Master, 1st Cohort — *Participant* Jul. – Aug. 2024

- Performed software fuzzing and vulnerability analysis.
- Discovered vulnerabilities in BandiView, Hauri ViRobot, and SGA ViRUSCHASER.

Stealien SSL Program, 3rd Cohort — *Participant* Oct. 2023 – Feb. 2024

- Analyzed WordPress plugin 1-days and found a 0-day arbitrary file upload in Icons Font Loader (CVE-2024-24714).
- Developed VulZap, a modular Python web vulnerability scanner combining endpoint

discovery (crawler and mitmproxy-based proxy), XSS/SQLi detection modules, and a MySQL-backed pipeline that tracks scan status and stores results.

KITRI WhiteHat School, 1st Cohort — *Participant* Sep. 2023 – May 2024

- Developed and trained a LLaMA-style language model with a custom SentencePiece tokenizer to generate SQLi/XSS payloads, validating generated SQLi payloads against DVWA.

Talks

KISA Hunting Master 2nd Cohort — *Special Lecture* Aug. 2025
Getting Started with Smart-Contract Audit Contests [PDF]

Blockchain Valley — *Security Team Workshop* Mar. – Jun. 2025
Web3 Security: Smart-Contract Threats & Hands-on Labs

27th Hacking Camp by PoC Security Aug. 2023
Hacking with AI: Try Web Hacking with Reinforcement Learning [PDF]

Disclosed Vulnerabilities

Anthropic Claude Code — Sandbox Bypass (High) [Write-up] HackerOne #3795252

Apple CoreMedia — Playback Denial of Service CVE-2025-24184

NASA CryptoLib — Heap-Based Buffer Overflow [Report] CVE-2025-30216

Pixar OpenUSD — Use-After-Free GHSA-58p5-r2f6-g2cj

PDFio — TTF Parser Denial of Service [Report] CVE-2024-42358

Toucan Protocol — Immunefi

- Medium *(private report)* Immunefi #40684
- Medium *(private report)* Immunefi #40596
- Low *(private report)* Immunefi #40497

BandiView

- JXR File Parser — Denial of Service CVE-2024-45870
- PSD File Parser — Denial of Service CVE-2024-45871
- PSD File Parser — Heap-Based Buffer Overflow CVE-2024-45872

WordPress Plugins

- Icons Font Loader — Arbitrary File Upload CVE-2024-24714
- Fast Custom Social Share by CodeBard — Stored XSS CVE-2023-48329
- Simple File List — Stored XSS CVE-2023-39924
- wpView — Stored XSS CVE-2023-33213

Hauri ViRobot — DLL Hijacking KVE-2024-0395

SGA ViRUSCHASER — DLL Hijacking KVE-2024-0410

Achievements & Awards

5th	Defense AI Cyber Security Hackathon	2026
10th	LakeCTF Finals 25-26 with CyKor	2026
1st	Cyber Security Hackathon with Boolean	2024

Finalist	CODEGATE Junior	2023
Finalist	WACON Junior with 1등하면전액기부팀	2023
Finalist	WhiteHat Contest Junior	2023
Finalist	CCE Junior	2023
1st	Intelligent Mobility SW Contest (AI) with 파송송	2023
1st	Chungcheong Youth Hacking Defense Contest with 1등 커트라인	2022
1st	AI Self-Directed Research Competition	2022
2nd	Korea Code Fair Hackathon with Krita	2022
3rd	e-ICON World Contest with Green Energy	2022
4th	KOSPO Web Security Competition with 걸어서 웹 속으로	2022
4th	K-ICT Data Creator Camp with Finder	2022